

UCI Census Income Classification — Day 1 Executive Summary

Author: Armish Iqbal | Web3Geeks AI/ML Internship | Repository: [internship_web3geeks](#)

1. PROBLEM FRAMING & SUCCESS METRIC

- Objective: Binary classification of individual annual income (>50K vs. <=50K) for targeted wealth advisory outreach.
- Class Base Rate: 7,841 positive instances (24.08%) out of 32,561 total records (severe class imbalance).
- Primary Metric (Precision / F1): Direct customer acquisition carries high outbound costs. False Positives (contacting <=50K earners) waste sales capacity. We prioritize Precision (74%+) and F1-Score over unweighted Accuracy.

2. REPRODUCIBLE DATA SPLITS (70 / 10 / 20)

- Train Set (70%): 22,792 samples (5,488 positive, 24.08%) | Dev Set (10%): 3,256 samples (784 positive, 24.08%)
- Hold-Out Test Set (20%): 6,513 samples (1,569 positive, 24.09%) | Stratified on target with fixed seed (random_state=42).
- Leakage Safeguard: Hold-out test set remains strictly isolated until final model evaluation to prevent data snooping.

3. BASELINE MODELS BENCHMARK (ON 6,513 HOLD-OUT TEST SET)

	Baseline Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
1. Majority-Class (<=		0.7591	0.0000	0.0000	0.0000	0.2409	0.2409
2. Education Rule (>		0.7367	0.4746	0.4879	0.4811	0.6582	0.3548
3. Required for Use		> 0.8200	> 0.7500	> 0.6000	> 0.6500	> 0.8500	> 0.6500

4. INITIAL ERROR ANALYSIS (FP = 847, FN = 804)

- False Positives (Predicted >50K, Actual <=50K - 847 cases): Young degree holders (<30 years) in entry-level roles, part-time professionals (<35 hrs/wk), and single non-family heads of households with zero capital gains.
- False Negatives (Predicted <=50K, Actual >50K - 804 cases): Experienced trade professionals (Craft-repair, Transport) without higher degrees earning high overtime (45-60+ hrs/wk), and individuals with substantial capital gains (> \$5,000).

5. DAY 2 FEATURE ENGINEERING ROADMAP & MODELING TARGET

- Skewed Numerics: Apply log1p transforms on 'capital_gain'/'capital_loss' + binary flags ('has_capital_gain').
- Missing Imputation: Explicit 'Unknown' imputation for workclass (1,836 missing) and occupation (1,843 missing).
- Seniority Interactions: Engineer (age * education_num) and (hours_per_week * age) interaction features.
- Marital Consolidation: Group 'Married-civ-spouse' (high earner indicator) vs. single/separated.
- Target Metric for Week: Optimize F1-Score & PR-AUC to push precision >75% and recall to 60-70%+ simultaneously.

Exploratory Data Analysis: Target Distribution, Demographics & Working Hours

